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Groundwater

A Critical Analysis of Transverse Dispersivity Field Data

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Abstract

Transverse dispersion, or tracer spreading orthogonal to the mean flow direction, which is relevant e.g, for quantifying biodegradation of contaminant plumes or mixing of reactive solutes, has been studied in the literature less than the longitudinal one. Inferring transverse dispersion coefficients from field experiments is a difficult and error-prone task, requiring a spatial resolution of solute plumes which is not easily achievable in applications. In absence of field data, it is a questionable common practice to set transverse dispersivities as a fraction of the longitudinal one, with the ratio 1/10 being the most prevalent. We collected estimates of field-scale transverse dispersivities from existing publications and explored possible scale relationships as guidance criteria for applications. Our investigation showed that a large number of estimates available in the literature are of low reliability and should be discarded from further analysis. The remaining reliable estimates are formation-specific, span three orders of magnitude and do not show any clear scale-dependence on the plume traveled distance. The ratios with the longitudinal dispersivity are also site specific and vary widely. The reliability of transverse dispersivities depends significantly on the type of field experiment and method of data analysis. In applications where transverse dispersion plays a significant role, inference of transverse dispersivities should be part of site characterization with the transverse dispersivity estimated as an independent parameter rather than related heuristically to longitudinal dispersivity.

Introduction

Analyzing, predicting, and controlling the fate of contaminants in the subsurface are key tasks for water quality management. Therefore, solute transport in groundwater is a subject of practical interest and its modeling has been a research topic in the last decades.

Traditionally, the advection-dispersion equation (ADE) was adopted in order to quantify solute spreading in porous media at pore scale (Bear 1972),

$$\frac{\partial C}{\partial t} + U\frac{\partial C}{\partial x_1} = D_{dL}\frac{\partial^2 C}{\partial x_1^2} + D_{dT}\frac{\partial^2 C}{\partial x_2^2} + D_{dV}\frac{\partial^2 C}{\partial x_3^2}. \quad (1)$$

Here $C(\mathbf{x}, t)$ is the resident solute concentration in space $\mathbf{x} = (x_1, x_2, x_3)$ and time t. Flow is of constant velocity U in the horizontal direction x_1 while D_{dL}, D_{dT}, and D_{dV} are the pore-scale dispersion coefficients in longitudinal, transverse horizontal, and transverse vertical directions, respectively.

Pore-scale dispersion is commonly parameterized following Scheidegger (1961): the dispersion coefficient is the sum of pore-diffusion D_m and velocity-proportional hydrodynamic dispersion. For typical values of U, the *Peclet numbers* $Pe = Ud/D_m$ (d is the pore diameter) are much larger than unity and transport is advection dominated. Consequently, hydrodynamic dispersion is the dominating mechanism and dispersivities $\alpha_{di} = D_{di}/U$ with $i \in \{L, T, V\}$ are approximately constant.

Laboratory experiments indicate that α_{dL} is of the order of the pore diameter d for homogeneous and isotropic media. It was found that $\alpha_{dT} = \alpha_{dV}$ is much smaller than α_{dL} by a factor of $5 - 40$ (see e.g., Dagan [1989], figures 2.10.4 & 2.10.5 for early experiments).

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While more recent laboratory experiments (see e.g., for an overview Klenk and Grathwohl [2002]) provided additional data, the pore-scale dispersivity values are generally much smaller than those pertaining to transport in aquifers at field-scale.

It is common in practice to quantify field scale flow and transport in aquifers by the same ADE (1). However, dispersion at the macro-scale is not the result of pore-scale processes but related to aquifer heterogeneity. The heterogeneity manifests in spatial variability of hydraulic conductivity K, which is characterized by scales much larger than pore scale. Thus, macro-dispersivities $D_i = \alpha_i U, \quad i \in \{L, T, V\}$, are by orders of magnitude larger than the pore scale dispersivities values based on laboratory investigations, particularly the one characterizing longitudinal spreading of solutes $\alpha_L \gg \alpha_{dL}$ (see e.g., for a recent overview Zech et al. [2015]). Similarly, transverse horizontal and vertical macrodispersivities are larger than their pore-scale counterparts (i.e., $\alpha_T > \alpha_{dT}$ and $\alpha_V > \alpha_{dV}$), though to a lesser extent than in the longitudinal direction, as for instance shown for the Bordon aquifer (Sudicky et al. 1983). Thus, the process equation ADE (1) is the same for transport at pore- and field scale but with dispersion parameters resulting from inherently different mechanisms.

A large body of literature is devoted to the modeling of field scale dispersion, primarily the longitudinal one. For the transverse dispersion of concern here, a lesser, but nevertheless significant body of numerical and theoretical works have aimed at providing an explanation to the mechanisms leading to observed values of macrodispersivities. Yet, no definite guidelines to be used by practitioners are available and it is common in applications to use ad hoc values of α_T as a fraction of α_L, similar to the laboratory data. However, there is no rational ground, either theoretical or experimental, to justify such an approach.

The theoretical research on aquifer transverse dispersion is still ongoing. It aims to explain the dispersivity values identified in controlled field experiments, like for example, Borden Site and Cape Cod, as well as to overcome the lack of prediction capability by existing theories, like the first order approximation in log-conductivity variance (see e.g., Attinger et al. [2004]). The main proposed mechanisms are (1) the interplay between large-scale advection, which in turn is ruled by the spatial structure of heterogeneous conductivity, and (2) pore-scale dispersion with primarily its transverse and vertical components (see e.g., Fiori [1996]; Berglund and Fiori [1997]). A few possible additional mechanisms have been considered, for example, time varying oscillations of flow (Bellin et al. 1996), small scale features of hydraulic conductivity, (Jankovic et al. 2009; Di Dato et al. 2016) or spatial variations of local anisotropy (Cirpka et al. 2015). However, the sole aim of the present paper is to analyze the available field data, toward providing practitioners with typical values of observed transverse macrodispersivities. Therefore, we do not further review here neither the literature on laboratory experiments nor the available numerical or theoretical studies.

The approach adopted here is the one followed in our work on longitudinal dispersion (Zech et al. 2015). The starting point is the compendium of field data of Gelhar et al. (1992), covering publications available at the time. Our first task, which is retaken here for transverse dispersion, was to assess the reliability of the data by analyzing carefully the original papers used by Gelhar et al. (1992). Zech et al. (2015) concluded that retaining the reliable data solely invalidates the so called scale effect by which α_L increases with the distance covered by solute plumes. This conclusion has a considerable impact on the assumed values of $\alpha_{T,V}$ by the rule of thumb of adopting a fixed ratio. Additionally, we expanded the data base by collecting and incorporating the new data available after the publication of Gelhar et al. (1992).

Similarly to the laboratory data, the number of publications presenting field data on transverse dispersion, is much smaller than those covering longitudinal spreading, and of lesser reliability. It is due primarily to the difficulty of measuring in detail the transverse concentration profiles required for inferring $\alpha_{T,V}$ with the aid of spatial moments. Nevertheless, knowledge of both α_L and $\alpha_{T,V}$ is important for analyzing and predicting reactive transport. In particular, mixing in conjunction with transverse dispersion on one hand and degradation on the other, determine to a large extent the length of reactive plumes (Liedl et al. 2005; Cirpka et al. 2006, 2012; Maier and Grathwohl 2006; Zarlenga and Fiori 2013). In such cases, accurate estimate of the transverse dispersion coefficient is a prerequisite for reliable groundwater risk assessment, remediation, and natural attenuation at a field site.

The procedure followed here is similar to the work of Zech et al. (2015). Section 2 contains the criteria for evaluating the reliability and (re-)evaluation of field data. Reliable field transverse dispersion coefficients are presented and discussed in Section 3 along with the ratios between the different dispersion coefficients. Section 4 provides conclusions and recommendations.

Transverse Dispersivity Literature Data

Reliability Criteria

In their compendium, Gelhar et al. (1992) classified both longitudinal and transverse macrodispersivities of high (R1), moderate (R2) or low (R3) reliability according to predefined criteria. Zech et al. (2015) updated these criteria for the longitudinal macrodispersivity. We adopt and refine here the criteria for reliability assessment to the specifics of field-scale transverse dispersivities.

We maintain the classification of high (R1), moderate (R2), low (R3) reliability for field measurements. Criteria are defined in Table 1 for experimental aspects (test type, source, tracer, and observations) and analysis. Details on every aspect are discussed in the Supporting Information. The reliability class of a dispersivity value is based on the reliability of the individual aspects. For a study to be of a

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Table 1
Reliability Criteria for Transverse Horizontal α_T and Transverse Vertical α_V Dispersion Coefficients[1]

Reliability	R1: High	R2: Moderate	R3: Low/exclusion
Test type	Regional flow with pulse/ step input	Forced gradient tests with advanced analysis methods	Forced gradient tests (push-pull; converging/ diverging well flow; dipole)
Source history and geometry	Tracer input well defined		Unknown quantity and duration; spatially distributed sources
Tracer	Conservative	Non-conservative effects accounted for	Non-conservative effects not accounted for
Observations[2]	3D monitoring; high tracer mass recovery; ergodicity in plume development	2D depth averaged (α_T); low tracer mass recovery	No measurements offside the main flowpath (α_T); no vertical concentration resolution (α_V); low resolution of measurements
Analysis	Spatial moments analysis of Fickian plume; 3D numerical model with fine grid resolution and adequate calibration	BTC analysis; type curve matching; numerical model with moderate grid resolution and/or simple calibration; steady state plume length analysis	Analysis not matched to flow configuration; numerical model insensitivity to dispersivity and/or no calibration; fixed ratio between longitudinal and transverse dispersivity

[1]The reliability of a study is associated to the lowest reliability throughout all aspects.
[2]Criteria specific for α_T and α_V, noted in brackets.

certain reliability (e.g., R2), all experimental aspects and the data interpretation fulfill the criteria of that class of reliability (e.g., R2 or higher).

Low reliability (R3) results led here to exclusion of the value for further analysis. Notice that macrodispersivities classified of low reliability (R3) may be perfectly legitimate in the context and in relation to the objective of the underlying study, whose main goal did not include identification of transverse dispersivity.

We summarize herein the main aspects of the reliability criteria, accounting for the differences between Gelhar et al. (1992) and Zech et al. (2015) for longitudinal dispersivity α_L, by:

- Transverse dispersivities determined from tracer tests in a radial flow configuration are considered as reliable (R2) only if the interpretation method is able to distinguish between the effect of transverse dispersion and flow nonuniformity. Otherwise they are considered lowly reliable, thus excluded.
- Initial transport conditions, such as source geometry and timing of the release, are often unknown or highly uncertain, resulting in unreliable estimates of the macrodispersivity. Consequently, cases in which the source input is not well documented, as it is most often the case of spill events, are excluded (R3).
- Quantifying plume spreading in the transverse direction requires that concentrations are measured in the direction normal to the plume trajectory, which can be achieved best by measurements within control planes. Thus, transverse dispersivities are excluded (R3) when obtained from experiments where concentrations were only taken along the plume trajectory. In addition, dense point sampling, spanning vertically the entire plume, including the plume fringes, is required to reliably estimate transverse vertical dispersivity α_V.
- Determining transverse dispersivities requires a two- or three-dimensional analysis. An estimate of α_T/V is considered reliable if determined independently from the longitudinal dispersivity α_L. Generally, that is the case for spatial moment analysis but often not in large-scale numerical models. Transverse dispersivities estimated as a fixed ratio of the longitudinal dispersivity are excluded (R3).
- Reliable estimates of transverse dispersivity are achieved by interpreting detailed field data with suitable models either analytical or numerical. A specific method, is the estimation of α_T/V from steady state plume length (Liedl et al. 2005; Cirpka et al. 2006, 2012; Maier and Grathwohl 2006; Zarlenga and Fiori 2014). Values of transverse dispersivity obtained in this way are evaluated as moderately reliable (R2) due to the two sources of uncertainty: first, the question whether the plume is indeed at steady state or not; second, the source strength, being an important factor in the steady state plume length equation, is often unknown. One way to overcome the latter uncertainty is to measure the decrease of electron donor fluxes in two subsequent control planes and to back out α_T from numerical modeling (Rügner et al. 2004; Maier and Grathwohl 2006). Biogeochemical transport modeling studies are evaluated according to the criteria for numerical models with additional attention to the reactive component.

Reevaluation of Literature Data

We conducted a detailed inspection of the original papers or reports on all experiments on transverse dispersivities (α_T and/or α_V) listed in Gelhar et al. (1992) applying the refined criteria for transverse dispersivity in Table 1. The final result of the inspection is summarized

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Table 2
Summary of Studies Reporting Reliable Transverse Dispersivities

Site Name, Country	Reference	L^1	$\alpha_L{}^2$	$\alpha_T{}^3$	α_V	$\sigma_Y^{2\,5}$	R^6
Bonnaud, France	Molinari and Peaudecerf (1977); Sauty (1977)	32.5	2.7	0.11	—	—	2
Borden (1), Canada	Freyberg (1986); Rajaram and Gelhar (1991)	90	0.5	0.05	0.0022	0.24	1
Borden (2), Canada	Sudicky et al. (1983); Gelhar et al. (1985)	11	0.34	0.084	0.0019	0.24	2
Cambridge, Canada	Robertson et al. (1991)	130	1	0.01	0.004	—	2
Cape Cod, Massachusetts, USA	Garabedian et al. (1991); Hess et al. (1992)	212	0.96	0.018	0.0015	0.24	1
Grenoble, France	Courtois et al. (2000)	45	7	0.2	—	1.21	2
Grindsted, Denmark	Petersen et al. (1998); Bjerg et al. (1995)	50	0.29	0.015	0.045	0.47	2
Hebei Aquifer, China	Yang et al. (2001)	15.5	1.72	0.0013	—	—	2
Jutland, Denmark	Jensen et al. (1993); Bjerg et al. (1992)	200	0.45	0.001	0.0005	0.37	2
Krauthausen, Germany	Vanderborght and Vereecken (2002); Vereecken et al. (2000)	123	3.64	0.02	—	1.08	1
Osterhofen, Germany	Maier and Grathwohl (2006)	450	—	—	0.032	—	2
Sjoelund, Denmark	Prommer et al. (2006); Tuxen et al. (2003)	50	—	—	0.005	0.56	2
Twin Lake/ Chalk River, Canada	Moltyaner and Killey (1988)	40	—	—	0.0014	—	2

[1]Observation scale L in [m]; equals the maximal plume travel distance.
[2]Longitudinal dispersivity value α_L in [m] from Zech et al. (2015).
[3]Transverse horizontal dispersivity value α_T in [m].
[4]Transverse vertical dispersivity value α_V in [m].
[5]Log-conductivity variance σ_Y^2 from geostatistical investigations (if available).
[6]Re-evaluated reliability R: 1—high; 2—moderate.

in Table 2 and visualized in Figure 1. Transverse dispersivities are displayed in log-log-scale vs. the observation scale which is the maximal plume travel distance of the tracer plume. A detailed discussion of the experimental conditions and the elements that lead to the classification of each experimental measure is given in the Supporting Information.

The reclassification resulted in a significant reduction of the number of reliable field data. Two values of α_T and/or α_V out of the four field experiments classified by Gelhar et al. (1992) as highly reliable were confirmed as R1: the values obtained from spatial moments analysis of tracer tests at the Borden site (Rajaram and Gelhar 1991) and at Cape Cod (Garabedian et al. 1991). Of the other two, one was reclassified as moderately reliable (Bonnaud, Sauty (1977)) and the other one was excluded (Palo Alto, Valocchi et al. [1981]). Two values of the originally six moderately reliable values were confirmed as R2, whereas the others were excluded. All of the 18 values classified of low reliability by Gelhar et al. (1992) are confirmed as R3. All the excluded dispersivity values fall into (at least) one of the following classes:

- The test type is insensitive to transverse dispersion and the assumption of a fixed ratio between longitudinal and transverse dispersivity is adopted. Studies excluded for this reason are the field experiments at Palo Alto (Valocchi et al. 1981), Campuget (Iris 1980), and Mobile (Sykes et al. 1982).
- The experimental setup is insensitive to transverse dispersivity. For instance Thorpe and Barry (1977) explicitly reported difficulties in determining reliable estimates of α_T and α_V with reported ranges varying over two orders of magnitude for the two field experiments in the Heretaunga Plains Aquifer.
- Case studies of contamination with unknown source distribution, fixed ratio, coarse grid resolution, no calibration, and model insensitive to transverse dispersivities are excluded. All studies classified by Gelhar et al. (1992) as lowly reliable fall into this category.

Additional Dispersivity Data

We performed an extensive literature review on field studies on subsurface transport published in the last 25 years to supplement the data published by Gelhar et al. (1992). This search started from the recent analysis conducted by Zech et al. (2015) on longitudinal dispersion. We further considered studies dealing with steady-state organic contaminant plumes which are usually insensitive to the value of α_L.

The literature review identified additional reliable transverse dispersivity values from eight field experiments: six values for α_T and five for α_V, with three experiments providing values for both. Only one α_T value was evaluated as highly reliable, all others as moderately reliable. A detailed discussion is provided in the *Supporting Information*.

Table 2 summarizes the resulting dataset of reliable transverse dispersivities. They are shown in Figure 2a.

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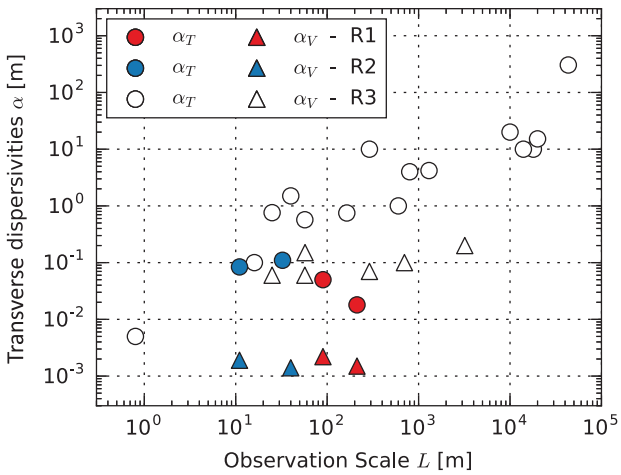

Figure 1. Transverse horizontal α_T (circles) and transverse vertical α_V (triangles) dispersivities vs. observation scale L of all experiments reported by Gelhar et al. (1992) with revised reliability: R1—high (red), R2—moderate (blue), R3—low/to exclude (white).

Longitudinal dispersivities are added to Table 2 for those sites where available. They are used for calculating the ratios between longitudinal and transverse dispersivities (Figure 3). For comparison as well as to give a visual impression of the magnitudes and scatter of the α_L values, we visualized them in Figure 2b in the context of all reliable field values reported by Zech et al. (2015).

Along with the dispersivity values, the log-conductivity variance σ_Y^2 is provided in Table 2 for those sites where it was available from literature studies. In all cases, σ_Y^2 refers to a geostatistical analysis of hydraulic conductivity measurements assuming a log-normal distribution, that is, σ_Y^2 is the variance of the normally distributed $Y = \ln K$. The log-conductivity variance value is indicative of the degree of heterogeneity of the aquifer and helps to interpret the estimated values of macrodispersivity.

Results

All reliable transverse dispersivity values are displayed in Figure 2a. The amount of data is limited to values from 12 different sites which confirms that transverse dispersion is much more difficult to measure than longitudinal dispersion, obvious when compared to the number of data points in Figure 2b, from almost four times as many different sites. The small amount of data on α_T and α_V suggests caution in drawing conclusions. However, this is the only data to provide reliable information on transverse dispersivity available from field experiments conducted in the past 50 years.

Reliable transverse dispersivity values in Figure 2a are scattered over about three orders of magnitude from $2 \cdot 10^{-4}$ m to $2 \cdot 10^{-1}$ m for a horizontal observation scale varying between 10 and 200 m. There are no apparent scale effects relative to the observation scale,

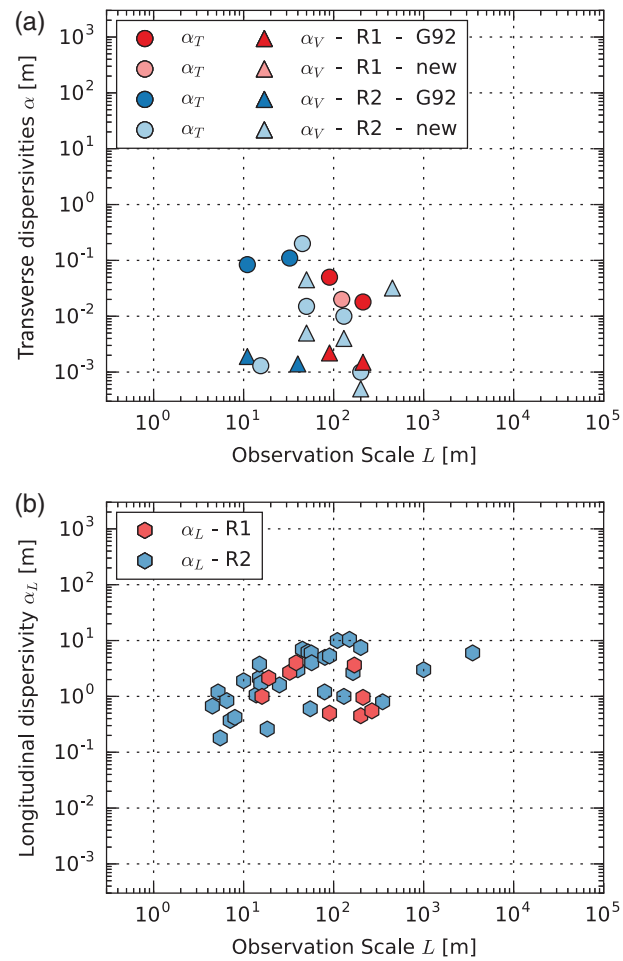

Figure 2. (a) (top) Reliable transverse horizontal α_T (circles) and transverse vertical α_V (triangles) dispersivities vs. observation scale, listed in Table 2. (b) (bottom) Reliable longitudinal dispersivities α_L vs. observation scale, from Zech et al. (2015). Values are marked according to high (R1, red) and moderate (R2, blue) reliability.

which is in line with theoretical results. The magnitude of transverse dispersivity values is much smaller than that of longitudinal dispersivity (Figure 2b); however, the scatter of values over several orders of magnitude is comparable.

Highly (R1) and moderately (R2) reliable dispersivities vary within similar ranges for both, Gelhar's and the new data. Figure 2a supports that transverse dispersivity is aquifer specific, given the large range of values and the lack of an evident scale effect. It shows that transverse dispersivity cannot be obtained from the travel distance, but requires a thorough aquifer characterization, similar to the longitudinal dispersivity (Figure 2b).

Figure 3 shows the ratios α_L/α_T, α_L/α_V and α_T/α_V for all field experiments reported in Table 2. The ratio α_L/α_T varies from 10 to 100 for data classified as R1, with the scatter ranging from 4 up to 1300 for R2. The ratio α_L/α_V varies over a similar range. Thus, transverse dispersivities cannot be obtained as a constant fraction of the longitudinal one, as is often done in applications. Instead, the three dispersivities should be evaluated independently by using field data.

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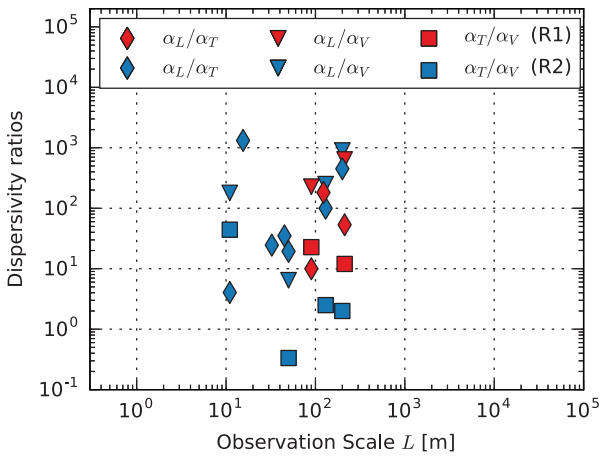

Figure 3. Ratios between longitudinal α_L, transverse horizontal α_T, and transverse vertical α_V dispersivities of high (R1, red) and moderate (R2, blue) reliability. The corresponding dispersivity values are listed in Table 2.

The ratio between vertical dispersivities, α_T/α_V, also varies over two order of magnitude from 1/3 to 44, mostly with α_V being one order of magnitude smaller then α_T. In some cases the ratio between both transverse dispersivities is larger than the ratio between transverse horizontal and longitudinal dispersivities, thus $\alpha_T/\alpha_V > \alpha_L/\alpha_T$, which may be indicative of stratification of the geological formation.

Conclusions and Recommendations

We assessed the reliability of transverse horizontal and transverse vertical dispersivities, α_T and α_V, available in literature, for field experiments conducted in unconsolidated media. The dataset comprises the data published in the compendium of Gelhar et al. (1992) and those collected from a literature review covering the last 25 years. All dispersivity values were classified into three categories of high (R1), moderate (R2), and low (R3) reliability, according to specific criteria (Table 1). Data of low reliability (R3) were excluded from the analysis.

The analysis of the available transverse dispersivity data showed that reliable values (R1 and R2) are site-specific and vary within about three orders of magnitude without any appreciable scaling with the plume travel distance or dimension of the investigated domain. Furthermore, some of the observed transverse dispersivities are rather small (of the order of millimeters), thereby suggesting that highly sensitive field observations are required to reliably infer them from field measurements. Such small values are comparable with those pertaining to effective pore-scale dispersion as, for example, the values identified by Fiori and Dagan (1999) for Cape Cod and Borden Site field experiments.

Transverse dispersion is a phenomenon whose full complexity is yet to be understood even after decades of research. The potential for new findings is still high in view of the limited amount of reliable field data. Although the paper addresses transport in unconsolidated aquifers solely, it is worthwhile to mention that Paster (2010) determined a rather small vertical dispersivity of $\alpha_V =$ 40 mm for a scale of $L = 30$ km in a highly conductive fractured aquifer. This value is in agreement with those represented in Figure 2a and supports the conclusion of lack of scale effect also for consolidated media. In contrast, Schwartz and Smith (1988) presented a model of fractured media showing that simple fracture patterns can lead to spreading with greater transverse than longitudinal dispersion.

The ratios between longitudinal and transverse dispersivities, α_L/α_T and α_L/α_V, obtained for field experiments where the three were evaluated independently, show a significant spread, in particular for α_L/α_T which varies between 4 and 1300. Practical conclusions are that vertical spreading, which is controlled by α_V, is small in natural formations and that the approach adopted by practitioners to take transverse dispersivity as a constant fraction of the longitudinal one is not warranted by field data and neither supported by theory.

Given the small values of α_T and α_V, an adequate aquifer's characterization and a careful identification of initial conditions are necessary requirements to obtain accurate estimates of transverse dispersivities. Site-speci fic investigations, although costly and time consuming, are still needed.

From an application perspective, the quality criteria presented in Table 1 are a useful guide to design field experiments and select appropriate data interpretation techniques. Characterization and modeling efforts should be coordinated to maximize the benefit of the investment. There is room for further development of efficient characterization techniques, as well as advanced models, to analyze groundwater transport like, for instance, methods of estimating transverse dispersivity from plume length observations (Liedl et al. 2005; Maier and Grathwohl 2006; Cirpka et al. 2012).

The present work should also serve as a cautionary note on the use of fixed ratios between longitudinal and transverse dispersivities, as they appear to be strongly aquifer dependent. It is highly recommended that a characterization plan includes the inference of the transverse dispersivities as a specific objective in applications in which they play a significant role.

Comparing the results for reliable field data of transverse (Figure 2a) vs. longitudinal (Figure 2b) dispersivity, we emphasize that the scarcity of reliable data is due to the difficulty of measuring transverse moments. Measuring transverse dispersion in the laboratory is carried out in sand boxes; it is somewhat more complicated than the standard column experiments employed for deriving the longitudinal dispersion coefficient. In contrast, measuring transverse dispersion in the field requires a three-dimensional observation network. Furthermore, scales and physics are often mixed for α_T: from molecular diffusion to plume meandering and from steady state to transient. This mix makes it difficult to distinguish dominant effects: diffusion, pore-scale dispersion, sub-plume variability, or macro-scale variability.

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This work casts doubt on the commonly used rule of thumb without providing an alternative. Unfortunately, there is no simple or inexpensive way of measuring transverse dispersion coefficients in the field. In the absence of site-specific data, the collection of transverse dispersivity values may be helpful in cases where a particular aquifer is geologically similar to one of those listed in Table 2. In any case, the values for α_T or α_V adopted in models are subject to large uncertainty. Relating transverse dispersivities to aquifer structural parameters is a challenging task requiring further investigation of theoretical and experimental nature.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article. Supporting Information is generally *not* peer reviewed.

Appendix S1. Supporting Information.

References

Attinger, S., M. Dentz, and W. Kinzelbach. 2004. Exact transverse macro dispersion coefficients for transport in heterogeneous porous media. *Stochastic Environmental Research and Risk Assessment* 18, no. 1: 9–15. https://doi.org/10.1007/s00477-003-0160-6

Bear, J. 1972. *Dynamics of Fluids in Porous Media*. New York: Elsevier.

Bellin, A., G. Dagan, and Y. Rubin. 1996. The impact of head gradient transients on transport in heterogeneous formations: Application to the Borden site. *Water Resources Research* 32, no. 9: 2705–2713. https://doi.org/10.1029/96WR01629

Berglund, S., and A. Fiori. 1997. Influence of transverse mixing on the breakthrough of sorbing solute in a heterogeneous aquifer. *Water Resources Research* 33, no. 3: 399–405. https://doi.org/10.1029/96WR03686

Bjerg, P.L., K. Hinsby, T.H. Christensen, and P. Gravesen. 1992. Spatial variability of hydraulic conductivity of an unconfined sandy aquifer determined by a mini slug test. *Journal of Hydrology* 136: 107–122. https://doi.org/10.1016/0022-1694(92)90007-I

Bjerg, P.L., K. Rugge, J. Pedersen, and T. Christensen. 1995. Distribution of redox-sensitive groundwater quality parameters downgradient of a landfill (Grindsted, Denmark). *Environmental Science & Technology* 29, no. 5: 1387–1394. https://doi.org/10.1021/es00005a035

Cirpka, O.A., A. Olsson, Q. Ju, M.A. Rahman, and P. Grathwohl. 2006. Determination of transverse dispersion coefficients from reactive plume lengths. *Groundwater* 44, no. 2: 212–221. https://doi.org/10.1111/j.1745-6584.2005.00124.x

Cirpka, O.A., M. Rolle, G. Chiogna, F.P.J. de Barros, and W. Nowak. 2012. Stochastic evaluation of mixing-controlled steady-state plume lengths in two-dimen sional heterogeneous domains. *Journal of Contaminant Hydrology* 138–139: 22–39. https://doi.org/10.1016/j.jconhyd.2012.05.007

Cirpka, O.A., G. Chiogna, M. Rolle, and A. Bellin. 2015. Transverse mixing in three-dimensional nonstationary anisotropic heterogeneous porous media. *Water Resources Research* 51, no. 1: 241–260. https://doi.org/10.1002/2014WR015331

Courtois, N., O. Gerbaux-Francois, C. Grenier, P. Maugis, E. Mouche, C. de Fouquet, P. Goblet, and E. Ledoux. 2000. Characterization of dispersion in an alluvial aquifer by tracing techniques and stochastic modelling. In *Calibration and Reliability in Groundwater Modelling*, Vol. 265, 84–89. Zürich: IAHS Press.

Dagan, G. 1989. *Flow and Transport on Porous Formations*. New York: Springer.

Di Dato, M., F.P.J. Barros, A. Fiori, and A. Bellin. 2016. Effects of the hydraulic conductivity microstructure on macrodispersivity. *Water Resources Research* 52, no. 9: 6818–6832. https://doi.org/10.1002/2016WR019086

Fiori, A. 1996. Finite Peclet extensions of Dagan's solutions to transport in anisotropic heterogeneous formations. *Water Resources Research* 32, no. 1: 193–198. https://doi.org/10.1029/95WR02768

Fiori, A., and G. Dagan. 1999. Concentration fluctuations in transport by groundwater: Comparison between theory and field experiments. *Water Resources Research* 35, no. 1: 105–112. https://doi.org/10.1029/98WR01862

Freyberg, D.L. 1986. A natural gradient experiment on solute transport in a sand aquifer: 2. Spatial moments and the advection and dispersion of nonreactive tracers. *Water Resources Research* 22, no. 13: 2031–2046. https://doi.org/10.1029/WR022i013p02031

Garabedian, S.P., D.R. LeBlanc, L.W. Gelhar, and M.A. Celia. 1991. Large-scale natural gradient tracer test in sand and gravel, Cape Cod, Massachusetts: 2. Analysis of spatial moments for a nonreactive tracer. *Water Resources Research* 27, no. 5: 911–924. https://doi.org/10.1029/91WR00242

Gelhar, L. W., A. Mantoglou, C. Welty, and K. R. Rehfeldt, A review of field-scale physical solute transport processes in saturated and unsaturated porous media, *Tech. Rep. EA-4190*, Elec. Power Res. Inst., Palo Alto, Calif., 1985.

Gelhar, L.W., C. Welty, and K.R. Rehfeldt. 1992. A critical review of data on field-scale dispersion in aquifers. *Water Resources Research* 28, no. 7: 1955–1974. https://doi.org/10.1029/92WR00607

Hess, K., S. Wolf, and M. Celia. 1992. Large-scale natural gradient tracer test in sand and gravel, Cape-Cod, Massachusetts.3. Hydraulic conductivity variability and calculated macrodispersivities. *Water Resources Research* 28, no. 8: 2011–2027. https://doi.org/10.1029/92WR00668

Iris, P., Contribution à l'étude de la valorisation énergétique des aquifères peu profonds expérience de stockage thermique en nappe phréatique, Ph.D. thesis, Univ. Pierre et Marie Curie et Ecole Natl. Super. des Mines, Paris, 1980.

Jankovic, I., D.R. Steward, R.J. Barnes, and G. Dagan. 2009. Is transverse macrodispersivity in three-dimensional groundwater transport equal to zero? A counterexample. *Water Resources Research* 45, no. 8: W08415. https://doi.org/10.1029/2009WR007741

Jensen, K.H., K. Bitsch, and P.L. Bjerg. 1993. Large-scale dispersion experiments in a sandy aquifer in Denmark: Observed tracer movements and numerical analyses. *Water Resources Research* 29, no. 3: 673–696. https://doi.org/10.1029/92WR02468

Klenk, I.D., and P. Grathwohl. 2002. Transverse vertical dispersion in groundwater and the capillary fringe. *Journal*

17456584, 2019, 4, Downloaded from https://ngwa.onlinelibrary.wiley.com/doi/10.1111/gwat.12838 by Universitatsbibliothek, Wiley Online Library on [02/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of Contaminant Hydrology 58: 111–128. https://doi.org/10.1016/S0169-7722(02)00011-6

Liedl, R., A.J. Valocchi, P. Dietrich, and P. Grathwohl. 2005. Finiteness of steady state plumes. *Water Resources Research* 41, no. 12: W12501. https://doi.org/10.1029/2005WR004000, wOS:000234297100001.

Maier, U., and P. Grathwohl. 2006. Numerical experiments and field results on the size of steady state plumes. *Journal of Contaminant Hydrology* 85: 33–52. https://doi.org/10.1016/j.jconhyd.2005.12.012

Molinari, J., and P. Peaudecerf. 1977. Essais conjoints en laboratoire et sur le terrain en vue d'une approach simplefiée de la prévision des propagations de substances miscibles dans les aquifères réels. In *Symposium on Hydrodynamic Diffusion and Dispersion in Porous Media*. Pavis, Italy: Int. Assoc. for Hydraul. Res.

Moltyaner, G.L., and R.W.D. Killey. 1988. Twin Lake Tracer Tests: Transverse dispersion. *Water Resources Research* 24, no. 10: 1628–1637. https://doi.org/10.1029/WR024i010p01628

Paster, A. 2010. Mixing between fresh and salt waters at aquifer regional scale and identification of transverse dispersivity. *Journal of Hydrology* 380: 36–44. https://doi.org/10.1016/j.jhydrol.2009.10.018

Petersen, M.J., P. Engesgaard, P.L. Bjerg, and K. Rügge. 1998. An anaerobic field injection experiment in a landfill leachate plume, Grindsted, Denmark: 1. Experimental setup, tracer movement, and fate of aromatic and chlorinated compounds. In *Groundwater Quality: Remediation and Protection*, 433–436. IAHS Publication No. 250.

Prommer, H., N. Tuxen, and P.L. Bjerg. 2006. Fringe-controlled natural attenuation of phenoxy acids in a landfill plume: Integration of field-scale processes by reactive transport modeling. *Environmental Science & Technology* 40, no. 15: 4732–4738. https://doi.org/10.1021/es0603002

Rajaram, H., and L.W. Gelhar. 1991. Three-dimensional spatial moments analysis of the Borden tracer test. *Water Resources Research* 27, no. 6: 1239–1251. https://doi.org/10.1029/91WR00326

Robertson, W.D., J.A. Cherry, and E.A. Sudicky. 1991. Groundwater contamination from two small septic systems on sand aquifers. *Groundwater* 29, no. 1: 82–92. https://doi.org/10.1111/j.1745-6584.1991.tb00500.x

Rügner, H., T. Holder, U. Maier, M. Bayer-Raich, P. Grathwohl, and G. Teutsch. 2004. Natural Attenuation-Untersuchungen "ehemalige Abfalldeponie Osterhofen." *Grundwasser* 9, no. 2: 98–108. https://doi.org/10.1007/s00767-004-0031-1

Sauty, J. P., Contribution à l'identification des paramètres de dispersion dans les aquifères par interprétation des expériments de traçage, Ph.D. thesis, Univ. Scientifique et Médicale et Institut Natl. Polytechnique de Grenoble, Grenoble, France, 1977.

Scheidegger, A. 1961. General theory of dispersion in porous media. *Journal of Geophysical Research* 66, no. 10: 3273–3278. https://doi.org/10.1029/JZ066i010p03273

Schwartz, F., and L. Smith. 1988. A continuum approach for modeling mass-transport in fractured media. *Water Resources Research* 24, no. 8: 1360–1372. https://doi.org/10.1029/WR024i008p01360

Sudicky, E.A., J.A. Cherry, and E.O. Frind. 1983. Migration of contaminants in groundwater at a landfill: A case study: 4. A natural-gradient dispersion test. *Journal of Hydrology* 63: 81–108. https://doi.org/10.1016/0022-1694(83)90224-X

Sykes, J.F., R.B. Lantz, S.B. Pahwa, and D.S. Ward. 1982. Numerical simulation of thermal energy storage experiment conducted by Auburn University. *Groundwater* 20, no. 5: 569–576. https://doi.org/10.1111/j.1745-6584.1982.tb01372.x

Thorpe, H. R., and B. J. Barry, Movement of Contaminants into and through the Heretaunga Plains Aquifer, *report*, N. Z. Minist. of Works and Dev., Wellington, 1977.

Tuxen, N., P. Ejlskov, H.J. Albrechtsen, L.A. Reitzel, J.K. Pedersen, and P.L. Bjerg. 2003. Application of natural attenuation to ground water contaminated by phenoxy acid herbicides at an old landfill in Sjoelund, Denmark. *Groundwater Monitoring and Remediation* 23, no. 4: 48–58. https://doi.org/10.1111/j.1745-6592.2003.tb00694.x

Valocchi, A., P. Roberts, G. Parks, and R. Street. 1981. Simulation of the transport of ion-exchanging solutes using laboratory-determined chemical-parameter values. *Groundwater* 19, no. 6: 600–607. https://doi.org/10.1111/j.1745-6584.1981.tb03514.x

Vanderborght, J., and H. Vereecken. 2002. Estimation of local scale dispersion from local breakthrough curves during a tracer test in a heterogeneous aquifer: The Lagrangian approach. *Journal of Contaminant Hydrology* 54: 141–171. https://doi.org/10.1016/S0169-7722(01)00143-7

Vereecken, H., U. Döring, H. Hardelauf, U. Jaekel, U. Hashagen, O. Neuendorf, H. Schwarze, and R. Seidemann. 2000. Analysis of solute transport in a heterogeneous aquifer: The Krauthausen field experiment. *Journal of Contaminant Hydrology* 45: 329–358. https://doi.org/10.1016/S0169-7722(00)00107-8

Yang, Y., X. Lin, T. Elliot, and R. Kalin. 2001. A natural-gradient field tracer test for evaluation of pollutant-transport parameters in a porous-medium aquifer. *Hydrogeology Journal* 9, no. 3: 313–320. https://doi.org/10.1007/s100400100127

Zarlenga, A., and A. Fiori. 2013. Steady plumes in heterogeneous porous formations: A stochastic Lagrangian approach. *Water Resources Research* 49, no. 2: 864–873. https://doi.org/10.1002/wrcr.20106

Zarlenga, A., and A. Fiori. 2014. Stochastic analytical modeling of the biodegradation of steady plumes. *Journal of Contaminant Hydrology* 157: 106–116. https://doi.org/10.1016/j.jconhyd.2013.11.003

Zech, A., S. Attinger, V. Cvetkovic, G. Dagan, P. Dietrich, A. Fiori, Y. Rubin, and G. Teutsch. 2015. Is unique scaling of aquifer macrodispersivity supported by field data? *Water Resources Research* 51, no. 9: 7662–7679. https://doi.org/10.1002/2015WR017220

17456584, 2019, 4, Downloaded from https://ngwa.onlinelibrary.wiley.com/doi/10.1111/gwat.12838 by Universitatsbibliothek, Wiley Online Library on [02/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License